

The pan-genome sheaf sees nothing: the recombination group of a pan-genome graph, and its torsion

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Abstract

A pan-genome graph is the pooled de Bruijn multigraph G of strains $H_1, \dots, H_p$ together with the strain walks, so the decomposition of the pooled flow into strains, which the phasing note had to count, is here given. The natural cellular sheaf on G — stalk the free abelian group on the strains through a vertex, restriction maps the projections — is the direct sum of the constant sheaves on the strain subgraphs G_i : its cohomology is $\mathbb{Z}^p$ and $\bigoplus_i H^1(G_i)$, and the integer cokernel of its sheaf Laplacian is $\bigoplus_i (\mathbb{Z} \oplus \mathcal{K}(|G_i|))$. It contains no pan-genomic information. What does is the *recombination group* $\mathcal{R} = Z_1(G) / \sum_i Z_1(G_i)$, the circulations of the pooled graph modulo those that live within single strains: the cycles that exist only by switching strain. The Mayer–Vietoris spectral sequence of the cover $\{G_i\}$ identifies $\mathcal{R}$ with the first homology of the nerve of intersection components, $C_q = \bigoplus_{|\sigma|=q+1} H_0(G_\sigma)$. For two strains $\mathcal{R}$ is free of rank $c(G_1 \cap G_2) - 1$ — the number of shared blocks minus one, which on a repeat-free reference with b bubbles is $b - 1$ and on ϕ X174, whose two-copy repeat glues shared blocks together, is smaller. For three or more strains $\mathcal{R}$ can have torsion: six synthetic strains whose intersection pattern is the six-vertex triangulation of the projective plane have $\mathcal{R} = \mathbb{Z}/2$, while the octahedron gives 0 and the seven-vertex torus $\mathbb{Z}^2$. A torsion class is a mosaic cycle twice of which, but not itself, is a sum of within-strain cycles; no single strain and no pair can see it. Six ϕ X174 isolates from GenBank, strand- and origin-normalised, give $\mathcal{R} = \mathbb{Z}^{38}$ with all 63 intersections non-empty and no torsion. Every group is computed twice, by a Smith form of the inclusion of the strain cycle lattices and from the nerve, and the two agree in all 28 pools. On complete *E. coli* chromosomes — where the cycle lattice is far out of reach but the nerve is not — $\mathcal{R}$ is free, and freely so at every prime at once: the Čech boundary d_2 reduces to the identity under ± 1 pivots, which certifies that all its elementary divisors are 1. Torsion needs a tight triangulated surface, and a real pan-genome nerve, whose strains share their core in tens of thousands of separate blocks, is the opposite shape. The certificate holds on every pool from two to eight chromosomes, up to $\mathcal{R} = \mathbb{Z}^{117\,848}$ on a nerve with 1.7 million two-cells, and does so even where d_2 has a large kernel; the rank of $\mathcal{R}$ saturates as strains are added, measuring the diversity of the set rather than its size.

Keywords. pan-genome graph, variation graph, cellular sheaf, sheaf Laplacian, cycle space, Mayer–Vietoris spectral sequence, nerve, recombination, torsion.

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1 What was asked, and what is true

The plan for this volume listed a note on “cellular sheaves on pan-genome graphs” and ranked it last, with the warning that the sheaf might turn out to be a relabelling of what the phasing note already had. This note carries out the construction and reports that the warning was right in one direction and wrong in another. The cellular sheaf that a pan-genome graph carries naturally is trivial in a precise sense (Proposition 2); but the question the sheaf was meant to ask — what does the pooled graph know that no single strain knows — has a clean answer that is not a relabelling of anything, and it comes with an integer invariant that can be torsion.

A pan-genome graph is a sequence graph together with the walks of the individual genomes through it [4, 3]. In the de Bruijn setting of these notes, that is the pooled multigraph $G = G_k(H_1 + \dots + H_p)$ of [1] with the decomposition $m = \sum_i \text{spec}(H_i)$ known. The phasing note counted the decompositions; here there is one, and the object of interest is the relation between the strain subgraphs G_i and their union. The cellular sheaf $\mathcal{F}$ that records, at each vertex and arc, which strains pass through it is the sheaf-theoretic encoding of that relation, and by the spectral theory of cellular sheaves [5] one would compute its cohomology and its sheaf Laplacian. Proposition 2 says both decompose strain by strain: $\mathcal{F}$ is the direct sum of the constant sheaves on the G_i , and everything one computes from it is a list of per-strain quantities. The sheaf sees no interaction because the projections that define it never mix strains.

The interaction lives one step further along. The circulation lattice $Z_1(G)$ of the pooled graph contains the circulation lattices $Z_1(G_i)$ of the strains, and the quotient

$$\mathcal{R}(H_1, \dots, H_p; k) = Z_1(G) / \sum_{i=1}^p Z_1(G_i)$$

is the group of cycles of the pan-genome that cannot be assembled from cycles of individual genomes — the cycles a walk can only close by switching from one strain to another at a shared k -mer, which is to say the recombination cycles. Section 3 computes it. The Mayer–Vietoris spectral sequence of the cover of G by the G_i identifies $\mathcal{R}$ with the first homology of a small chain complex built from the connected components of the intersections $G_{i_0} \cap \dots \cap G_{i_q}$ (Theorem 4). Two consequences follow. For two strains, $\mathcal{R}$ is free of rank one less than the number of shared blocks (Corollary 5). For three or more, $\mathcal{R}$ can have torsion, and Proposition 6 exhibits six strains with $\mathcal{R} = \mathbb{Z}/2$. Section 4 verifies all of it, and computes $\mathcal{R}$ for six deposited ϕ X174 genomes.

Throughout, strains are circular sequences over $\{A, C, G, T\}$, k is fixed, and the pooled graph is unitig-compacted with the branch criterion of [1] (a k -mer is a vertex if it has two distinct successors or two distinct predecessors in the pool); a *content* is a unitig string, A the set of contents, and $Z_1(G) \subset \mathbb{Z}^A$ the circulation lattice. All computations are exact; the script `pangenome_sheaf.py` reproduces every number below and prints a five-line pass/fail battery.

2 The pan-genome sheaf

Definition 1. Let $G_i \subseteq G$ be the subgraph of contents and vertices used by strain i , and for a vertex or content x let $S(x) = \{i : x \in G_i\}$. The *pan-genome sheaf* $\mathcal{F}$ has stalks $\mathcal{F}(x) = \mathbb{Z}^{S(x)}$ and, for a content a with endpoint v , the restriction map $\mathcal{F}(v) \rightarrow \mathcal{F}(a)$ the coordinate projection $\mathbb{Z}^{S(v)} \rightarrow \mathbb{Z}^{S(a)}$ (note $S(a) \subseteq S(v)$). Its coboundary $\delta: \bigoplus_v \mathcal{F}(v) \rightarrow \bigoplus_a \mathcal{F}(a)$ is $(\delta x)_a = x_{\text{head}(a)}|_{S(a)} - x_{\text{tail}(a)}|_{S(a)}$, and its sheaf Laplacian is $L_{\mathcal{F}} = \delta^T \delta$.

Proposition 2. $\mathcal{F} \cong \bigoplus_{i=1}^p \mathbb{Z}_{G_i}$, the direct sum of the constant sheaves $\mathbb{Z}$ supported on the strain subgraphs. Consequently $H^0(G; \mathcal{F}) \cong \mathbb{Z}^p$, $H^1(G; \mathcal{F}) \cong \bigoplus_i H^1(G_i; \mathbb{Z})$, $L_{\mathcal{F}}$ is block diagonal with blocks the graph Laplacians of the undirected shadows $|G_i|$, and

$$\text{coker } L_{\mathcal{F}} \cong \bigoplus_{i=1}^p (\mathbb{Z} \oplus \mathcal{K}(|G_i|)),$$

$\mathcal{K}(|G_i|)$ the critical group of the undirected multigraph $|G_i|$.

Proof. The coordinate i of every stalk is present exactly on the cells of G_i , and the restriction maps are the identity on that coordinate wherever both cells carry it and zero otherwise; that is the constant sheaf $\mathbb{Z}$ on the subcomplex G_i , extended by zero. Direct sums of sheaves have direct sums of cochain complexes, so cohomology and Laplacian decompose; the cochain complex of $\mathbb{Z}_{G_i}$ is the cellular cochain complex of the graph G_i , whose H^0 is $\mathbb{Z}$ for connected G_i (a strain walk is connected) and whose Laplacian is the graph Laplacian of $|G_i|$, with cokernel $\mathbb{Z} \oplus \mathcal{K}(|G_i|)$. $\square$

Check (F) computes $\text{coker } L_{\mathcal{F}}$ for a diploid with three substitutions and for three strains, obtaining $\mathbb{Z}/38 \oplus \mathbb{Z}/38 \oplus \mathbb{Z}^2$ and $(\mathbb{Z}/84)^3 \oplus \mathbb{Z}^3$ respectively, and confirms the block decomposition by a Smith form of the assembled Laplacian. The numbers 38 and 84 are the critical groups of the strains' own shadows; they would be the same if the strains were sequenced separately. There is nothing pan-genomic in the sheaf, and no choice of stalk dimensions rescues this as long as the restriction maps are projections: a sheaf whose restriction maps never mix coordinates is a direct sum of coordinate sheaves.

3 The recombination group

Definition 3. $\mathcal{R} = Z_1(G) / \sum_i Z_1(G_i)$, where $Z_1(G_i) = Z_1(G) \cap \mathbb{Z}^{A_i}$ is the circulation lattice of the subgraph G_i . For a subset $\sigma \subseteq \{1, \dots, p\}$ write $G_\sigma = \bigcap_{i \in \sigma} G_i$ and let $C_q = \bigoplus_{|\sigma|=q+1} H_0(G_\sigma)$, the free abelian group on the connected components of the $(q+1)$ -fold intersections, with the Čech differential $\partial(\sigma, c) = \sum_j (-1)^j (\sigma \setminus \sigma_j, c')$, c' the component of $G_{\sigma \setminus \sigma_j}$ containing c . We call $(C_\bullet, \partial)$ the *nerve of components* $\mathcal{N}$.

Theorem 4. $\mathcal{R} \cong H_1(\mathcal{N})$.

Proof. The subgraphs G_i are subcomplexes covering the one-dimensional CW complex G , so the Mayer–Vietoris spectral sequence of the cover [2, VII.4] applies: $E_{q,r}^1 = \bigoplus_{|\sigma|=q+1} H_r(G_\sigma) \Rightarrow H_{q+r}(G)$, with d^1 the Čech differential. Since intersections of graphs are graphs, $H_r(G_\sigma) = 0$ for $r \geq 2$, and the E^2 page has two rows, $E_{q,0}^2 = H_q(\mathcal{N})$ and $E_{q,1}^2 = H_q(\bigoplus_\sigma H_1(G_\sigma))$. The only differentials that can touch total degree one are $d^2: E_{2,0}^2 \rightarrow E_{0,1}^2$ and the vanishing $d^2: E_{1,0}^2 \rightarrow E_{-1,1}^2$, so $E_{1,0}^\infty = E_{1,0}^2 = H_1(\mathcal{N})$ and $E_{0,1}^\infty = E_{0,1}^2 / d^2(E_{2,0}^2)$. The filtration of $H_1(G)$ has $F_0 = E_{0,1}^\infty$ equal to the image of $\bigoplus_i H_1(G_i) \rightarrow H_1(G)$ (the edge homomorphism) and $H_1(G)/F_0 \cong E_{1,0}^\infty$. For a graph $H_1 = Z_1$, and the image of $\bigoplus_i Z_1(G_i)$ in $Z_1(G)$ is $\sum_i Z_1(G_i)$. $\square$

Corollary 5 (Two strains). *For $p = 2$ with G_1, G_2 connected and $G_1 \cap G_2 \neq \emptyset$, $\mathcal{R}$ is free of rank $c(G_1 \cap G_2) - 1$, the number of connected components of the shared subgraph minus one.*

Proof. $\mathcal{N}$ has $C_0 = \mathbb{Z}^2$, $C_1 = \mathbb{Z}^{c(G_{12})}$ and $C_2 = 0$; ∂ sends every component to $e_2 - e_1$, so $H_1(\mathcal{N}) = \ker \partial$ is the sublattice of coefficient sum zero. $\square$

On a repeat-free reference with b substitution sites in distinct $(k+1)$ -windows, the shared subgraph is the b stretches between the bubbles and $\mathcal{R} \cong \mathbb{Z}^{b-1}$. When the reference has a repeat the shared stretches that contain copies of the same repeated k -mer are joined through it and $c(G_1 \cap G_2)$ drops: for ϕ X174 at $k = 12$, whose two repeated 12-mers a, b each occur twice, five substitutions leave three shared components and $\mathcal{R} = \mathbb{Z}^2$, three substitutions leave two and $\mathcal{R} = \mathbb{Z}$ (Table 1). A repeat inside a strain thus reduces the number of independent recombination cycles between strains: the cycle through the repeat is already a within-strain cycle.

Proposition 6 (Torsion). *Let T be a triangulated closed surface with vertex set $\{1, \dots, p\}$, and construct strains as follows: a random segment F_τ for each face τ and E_ε for each edge ε ; strain i reads, for each edge $\varepsilon = ij$ at i with faces τ, τ' , the word $F_\tau E_\varepsilon F_{\tau'} S F_{\tau'} E_\varepsilon F_\tau S'$ with unique spacers S, S' . Then for k larger than any accidental shared k -mer the nerve of components is the simplicial chain complex of T , and $\mathcal{R} \cong H_1(T; \mathbb{Z})$. In particular the six-vertex triangulation of the projective plane gives $\mathcal{R} = \mathbb{Z}/2$, the octahedron gives 0, and the seven-vertex torus gives $\mathbb{Z}^2$.*

Proof. $G_i \cap G_j$ for an edge ij consists of E_{ij} , the two face segments at ij , and the four junction arcs between them, which both strains contain because each reads the two junction directions; it is connected. For a face ijk , $G_i \cap G_j \cap G_k = F_{ijk}$, a segment; for a non-face triple and for all larger subsets the intersection is empty, since every segment is shared by the vertices of exactly one simplex and every junction arc by exactly two strains. So $C_\bullet$ is the chain complex of T and Theorem 4 applies. $H_1(\mathbb{RP}^2; \mathbb{Z}) = \mathbb{Z}/2$, $H_1(S^2) = 0$, $H_1(T^2) = \mathbb{Z}^2$. $\square$

The torsion class is concrete. In the projective-plane pool there is a circulation z of the pooled graph such that $2z$ is a sum of cycles each lying within a single strain but z itself is not; z is a mosaic that closes up only by using every strain, and its double decomposes. No pair of strains detects it, because for pairs $\mathcal{R}$ is free (Corollary 5); no strain detects it, because within a strain there is nothing to detect. It is an invariant of the pan-genome as a whole.

4 Verification and data

Every group is computed twice: as the cokernel, via Smith normal form, of the matrix whose columns are fundamental cycles of the subgraphs G_i expressed in $\mathbb{Z}^A$ (torsion of $\mathbb{Z}^A / \sum \mathbb{Z}_1(G_i)$ equals that of $\mathcal{R}$ because $\mathbb{Z}_1(G)$ is saturated in $\mathbb{Z}^A$), and as $H_1(\mathcal{N})$ from the component structure of the intersections. The two agree in every one of the 28 pools, which is check (T) and part of the others.

The checks. In `pangenome_sheaf.py`: (F) the sheaf Laplacian's cokernel is the direct sum of the strains' $\mathbb{Z} \oplus \mathcal{K}(|G_i|)$; (B) two strains, eight pools, $\mathcal{R}$ free of rank $c(G_1 \cap G_2) - 1$; (S) the three surface pools, with the designed nerve; (T) twelve random three-strain pools from a shared segment library, Smith form and nerve agreeing, no torsion; (D) the six isolates. All pass in seven seconds. In `real_pangenome.py`, for Section 5: (A) the hashed branch criterion against brute force on random 20 kbp fragment pools; (V) the $\mathbb{F}_p$ detector against the three surface pools, where it returns the $\mathbb{Z}/2$ of the projective plane; (E) the complete chromosomes of Table 4.

5 A real pan-genome

Theorem 4 replaces a lattice quotient by the homology of the nerve, and the nerve needs only the *connected components* of the shared subgraphs. Components are cheap: the recombination

pool ($k = 12$)	$ V $	$ A $	$\text{rank } Z_1(G)$	$\text{rank } \sum Z_1(G_i)$	$c(G_1 \cap G_2)$	$\mathcal{R}$
ϕX174 , 1 substitution	4	7	4	4	1	0
ϕX174 , 2 substitutions	6	10	5	4	2	$\mathbb{Z}$
ϕX174 , 3 substitutions	8	13	6	5	2	$\mathbb{Z}$
ϕX174 , 5 substitutions	12	19	8	6	3	$\mathbb{Z}^2$
ϕX174 , sites 600, 606 linked +1	6	10	5	4	2	$\mathbb{Z}$
repeat-free reference, 2 sites	4	6	3	2	2	$\mathbb{Z}$
repeat-free reference, 3 sites	6	9	4	2	3	$\mathbb{Z}^2$
repeat-free reference, 4 sites	8	12	5	2	4	$\mathbb{Z}^3$

Table 1: Two strains. $\mathcal{R}$ is free of rank $c(G_1 \cap G_2) - 1$ in every row; on the repeat-free reference (a random 3000-mer) that is the number of bubbles minus one, on ϕX174 the repeat glues shared stretches and lowers it. Check (B).

surface pool ($k = 16$)	strains	$ V $	$ A $	$\text{rank } Z_1(G)$	$\text{rank } \sum Z_1(G_i)$	$\mathcal{R} = H_1(\mathcal{N})$
octahedron (S^2)	6	114	190	77	77	0
seven-vertex torus (T^2)	7	186	319	134	132	$\mathbb{Z}^2$
hemi-icosahedron ($\mathbb{RP}^2$)	6	138	233	96	96	$\mathbb{Z}/2$

Table 2: Proposition 6. In each pool the nerve of components is verified to be exactly the designed triangulation (one component per simplex, none elsewhere). At $k = 12$ the $\approx 10^4$ random 12-mers of a pool produce accidental shared k -mers and spurious components; $k = 16$ removes them. Check (S).

group is therefore computable on complete bacterial chromosomes, where $\mathcal{R}$ itself, as a quotient of a cycle lattice of rank 10^4 or more, is not. This section carries that out.

The pool. Complete *Escherichia coli* chromosomes were taken from GenBank and brought to the strand of K-12 MG1655 by k -mer overlap; a circular genome’s k -mer set is rotation invariant, so only the strand has to be fixed. The pooled graph is compacted by the criterion of [1]: a k -mer is a vertex when it has two distinct successors or two distinct predecessors *in the pool*. Branch vertices are found by hashing on the pairs (k-mer, next base) and (k-mer, previous base) and then re-tested on the exact strings, so a collision can only propose a candidate that the string pass discards; check (A) audits the whole criterion against brute force on random 20 kbp fragment pools.

A certificate, not a sample. Torsion is a statement about every prime, and one cannot test them all. Two devices avoid having to.

Proposition 7 (freeness certificate). *Suppose the Čech boundary $d_2: C_2 \rightarrow C_1$ can be reduced by integer elimination in which every pivot is ± 1 . Then all its elementary divisors are 1, $\text{im } d_2$ is a direct summand of C_1 , and $\mathcal{R} = \ker d_1 / \text{im } d_2$ is free.*

Proof. A ± 1 pivot is a unimodular row and column operation, so a complete reduction exhibits d_2 as $\begin{pmatrix} I \\ 0 \end{pmatrix}$ up to unimodular change of basis; its elementary divisors are all 1 and $C_1 / \text{im } d_2$ is free. A finitely generated submodule of a finitely generated free $\mathbb{Z}$ -module is free, and $\mathcal{R} = \ker d_1 / \text{im } d_2 \subseteq C_1 / \text{im } d_2$. $\square$

	accession	substitutions after normalisation					
1	LC786485.1	0	5	27	26	24	33
2	NC_001422.1	5	0	26	25	23	32
3	MH378443.1	27	26	0	19	24	48
4	MH378442.1	26	25	19	0	22	47
5	MH378441.1	24	23	24	22	0	45
6	JQ764988.2	33	32	48	47	45	0

Table 3: Six ϕ X174 genomes of length 5386 from GenBank, brought to the strand and origin of NC_001422.1 (one was deposited rotated). Pooled at $k = 12$: $|V| = 104$, $|A| = 158$, $\text{rank } Z_1(G) = 55$, $\text{rank } \sum Z_1(G_i) = 17$, $\mathcal{R} = \mathbb{Z}^{38}$, no torsion; all 63 intersections are non-empty, with 4 to 27 components for pairs and 41 for the full intersection. Check (D).

strains ($k = 31$)	branch vtx	contents	C_1	C_2	$\text{rank } d_2$	$\mathcal{R}$	certified
MG1655, DH10B	1 509	2 343	86	0	0	$\mathbb{Z}^{85}$	yes
+ O157:H7 Sakai	63 245	95 168	58 454	28 820	28 820	$\mathbb{Z}^{29\,632}$	yes
+ CFT073	137 375	207 028	166 011	134 401	99 561	$\mathbb{Z}^{66\,447}$	yes
+ IAI39	180 256	271 937	260 369	277 947	172 507	$\mathbb{Z}^{87\,858}$	yes
+ S88	214 033	323 073	409 007	588 230	304 507	$\mathbb{Z}^{104\,495}$	yes
+ 536	237 525	358 660	575 831	1 053 725	459 769	$\mathbb{Z}^{116\,056}$	yes
+ UTI89	241 592	364 819	744 598	1 683 865	626 743	$\mathbb{Z}^{117\,848}$	yes

Table 4: The recombination group of eight complete *E. coli* chromosomes, about 40 Mbp, each row adding one strain to the one above. In every row the unit-pivot reduction of d_2 completes, so by Proposition 7 $\mathcal{R}$ is free at every prime, and the rank, being $\dim C_1 - \text{rank } d_1 - \text{rank } d_2$ with both ranks computed by unimodular integer elimination, is exact. The two K-12 strains share 86 blocks and give $\text{rank } \mathcal{R} = 86 - 1$, which is Corollary 5 on real data. From the fourth strain on d_2 is far from injective — at eight strains 626 743 of 1 683 865 — which costs the certificate nothing: a kernel makes columns vanish during the reduction, it does not stop the surviving pivots being units. Check (E).

Remark 8 (a torsion detector without a Smith form). When the reduction does not complete, the prime-by-prime test is still cheap. $H_0(\mathcal{N})$ is free, so the universal coefficient theorem gives $\dim_{\mathbb{F}_p} H_1(\mathcal{N}; \mathbb{F}_p) = \text{rank } \mathcal{R} + t_p(\mathcal{R})$ with t_p the p -torsion rank: any excess of the $\mathbb{F}_p$ dimension over the rational rank is p -torsion, and both are ranks of sparse matrices. Check (V) runs this detector on the three surface pools of Table 2, where the answer is known exactly, and it does return the $\mathbb{Z}/2$ of the projective plane — so a null result on real data is a result and not a blind spot.

The answer. $\mathcal{R}$ is free, with a certificate, on every pool from two to eight complete chromosomes (Table 4). This settles the first open question of an earlier version of this note for those strains, and the reason is visible in the same table. A torsion class needs a tight two-dimensional cycle: the projective-plane pool of Proposition 6 has exactly one component per simplex, so its nerve is a triangulated surface. A real pan-genome nerve is the opposite shape. Each pair of divergent strains shares its core genome in tens of thousands of blocks — 86 for two K-12 strains, 28 817 and 29 551 for K-12 against O157:H7 — and each triple intersection is more fragmented still, so the nerve is wide and sparse, every elementary divisor of d_2 is 1, and H_1 is a large free group. The

synthetic construction of Section 3 was necessary precisely because real data does not produce that shape, and the honest reading of Proposition 6 is that torsion is realisable in principle rather than observed in *E. coli*.

The rank saturates. One thing in Table 4 is not about torsion. The successive increments of rank $\mathcal{R}$ are 29 547, 36 815, 21 411, 16 637, 11 561, 1 792: after the third strain each new chromosome contributes less than the one before, and the eighth contributes an order of magnitude less than the fourth, on a nerve whose two-cells meanwhile grew to 1.7 million. The pooled graph agrees — the last strain adds 4 067 branch vertices where the fifth added 42 881. The ordering is not neutral: the last four additions (CFT073, S88, 536, UTI89, with IAI39 between them) are extraintestinal pathogenic strains, so each arrives with most of its mosaic structure already represented, and the two K-12 strains at the top are a pair, not a sample. Read as an observation on these eight genomes rather than a law, it says that rank $\mathcal{R}$ is a measure of the *diversity* of a strain set and not of its size, which is what a count of independent strain-switching cycles ought to be.

Remark 9 (a failed certificate is a statement about the elimination). Proposition 7 is one-sided: it certifies freeness when the reduction completes and says nothing when it does not, because d_2 may have all its elementary divisors equal to one with no ± 1 pivot surviving to the end of one particular elimination. That is not a theoretical caveat. An earlier version of this computation chose each pivot by scanning the 64 shortest live columns, and stalled at four strains with 70 374 columns left and at five with 190 814, from which we could conclude only — via Remark 8 — that there is no 2- and no 3-torsion. Taking instead the globally shortest column, and within it the row meeting fewest columns, produces less fill-in, the entries never leave $\{0, \pm 1\}$, and every column is eliminated: those two rows of Table 4 became certificates, and the three below them became feasible at all. The ranks 99 561 and 172 507 are the same under both orderings, which is what says the two runs compute the same map. Two consequences for the arithmetic. Because the unit-pivot phase is a unimodular column reduction over $\mathbb{Z}$ it is valid modulo every prime with its pivots still units, so when it *does* stall, $\text{rank}_{\mathbb{F}_p} d_2$ is the number of pivots plus the rank of the residual block alone; and the ordering is worth more than the arithmetic, the five-strain reduction taking 4 seconds where the scan took 14 080.

What the single-strand convention costs here. Of the eight strains, IAI39 matched the reference on 2331 of its sampled 31-mers forward against 1234 reverse, where every other strain ran between 6835 against 234 and 2970 against 109. That is a large inversion, and to the single-strand pooled graph an inverted segment is strain-specific sequence: real homology is lost and the nerve is cut into more components than it should be. The rank of $\mathcal{R}$ reported here is therefore an overcount for divergent strains, in a way a bidirected pooled graph would repair. The freeness is not affected, being a statement about the shape of the nerve.

6 Scope, provenance, and what is open

Scope. Proposition 2, Theorem 4, Corollary 5 and Proposition 7 hold for arbitrary strains; Proposition 6 is a construction; Section 5 is a computation on named genomes and claims nothing beyond them. The pooled graph is single-stranded, as in [1]; the reverse-complement version would pool D_k ’s and is not attempted. “Recombination” is used in the graph sense — a cycle

that switches strain — and no claim is made about the biological mechanism that produced any particular mosaic.

Provenance. Cellular sheaves on graphs and their Laplacians are [5]; pan-genome and variation graphs are [4, 3], and the bubble/snarl decomposition of such graphs, which is the combinatorial shadow of the components of pairwise intersections, is [6]. The Mayer–Vietoris spectral sequence for a cover by subcomplexes is standard, e.g. [2, VII.4]; Theorem 4 is its two-row case read off for graphs. We are not aware of prior work defining the quotient $Z_1(G)/\sum Z_1(G_i)$ for a pan-genome graph or observing that it can have torsion; a targeted search found none.

Open. *First*, torsion in a real pan-genome. Section 5 answers this for the *E. coli* chromosomes pooled there — $\mathcal{R}$ is free, with a certificate — and explains why: the nerve of a real pan-genome is wide and sparse, nothing like the triangulated surface a torsion class needs. What is open is whether any real pan-genome has one. A plausible place to look is a set of strains whose shared blocks are *few and large*, so that the nerve is small enough to close up: closely related strains with a handful of recombination breakpoints, rather than the divergent ones used here. *Second*, the relation of $H_1(\mathcal{N})$ to the snarl tree of [6], which organises the pairwise components hierarchically. *Third*, a bidirected pooled graph, so that an inversion is homology rather than strain-specific sequence; by the last paragraph of Section 5 this changes the rank of $\mathcal{R}$ for divergent strains, and Bio 7’s anti-automorphism theorem says the repair cannot be a quotient by the strand involution. *Fourth*, a sheaf that does see interaction: the restriction maps would have to mix strains, and the natural candidate is the sheaf of local decompositions of the pooled flow, whose stalk at a bubble is the homogeneous space of [1]; it is not a sheaf of abelian groups.

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